

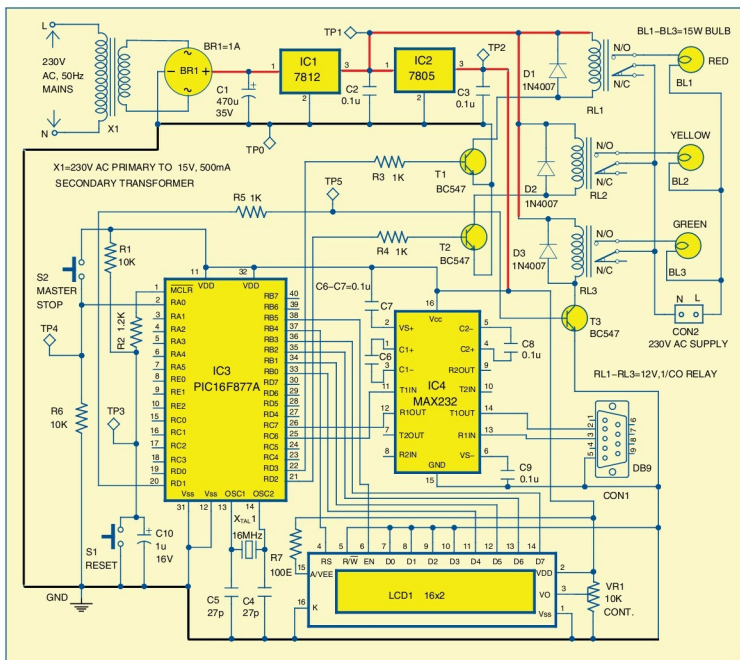


Fig. 2: Circuit diagram of the programmable interval timer

over, yellow bulb starts flickering to indicate that the grace period has started.

When the grace period is over, yellow bulb goes off and the red bulb glows; and immediately the in-built speaker of the PC produces an alert sound, indicating the grace period is over. If the item does not stop before the red bulb glows, the item being presented gets disqualified.

For the next team or contestant, you have to press reset button S1 to repeat the process.

Power supply. The power supply section comprises a 230V AC primary to 15V, 500mA secondary transformer (X1), bridge rectifier module (BR1),

Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	5V
TP3	Low when S1 is pressed
TP4	High when S2 is pressed
TP5	High when start button is pressed

regulator ICs 7812 and 7805. The output of bridge rectifier is filtered by capacitor C1 and fed to input of 12V regulator IC1. The 12V drives the three relays RL1 through RL3 and is also fed to input of 5V regulator IC2.

The 5V supply drives the microcontroller (IC3), LCD1 and MAX232 (IC4). Resistors R1 and R2 and capacitor C10 form power-on reset circuitry of PIC16F877A microcontroller.

Software

The software part includes two programs, that is, user-interface program and firmware for the microcontroller. Along with these programs, there is an MS Access database file (.mdb) where the user has already entered the minimum, maximum and grace timings (refer Fig. 3). User can change these timings as per requirement, as explained in the next section.